

6.9 Continuous-time Markov chains

Let $X = \{X(t) : t \geq 0\}$ be a family of random variables taking values in some countable state space S and indexed by the half-line $[0, \infty)$. As before, we shall assume that S is a subset of the integers.

(A) Markov property.

(1) Definition. The process X satisfies the **Markov property** if

$$\mathbb{P}(X(t_n) = j \mid X(t_1) = i_1, \dots, X(t_{n-1}) = i_{n-1}) = \mathbb{P}(X(t_n) = j \mid X(t_{n-1}) = i_{n-1})$$

for all $j, i_1, \dots, i_{n-1} \in S$ and any sequence $t_1 < t_2 < \dots < t_n$ of times.

There is some complexity in the general theory of processes with the Markov property when S is infinite (the finite case is more straightforward). The way to handle this is too difficult to describe in detail here, and the reader should look elsewhere (see Chung 1960, or Freedman 1971 for example). We follow a different route by requiring a degree of smoothness of the sample paths of X , namely right-continuity (see (6.8.23) for the definition of right-continuity in this context).

(2) Definition. The process X is a **(continuous-time) Markov chain** if it is right-continuous and satisfies the Markov property.

If X is a continuous-time Markov chain, any homogeneous discretization of time yields a discrete-time Markov chain, as follows. Let $h > 0$, and define $Z_n = X(nh)$. It is an immediate consequence of the Markov property (1) that $Z = \{Z_n : n \geq 0\}$ is a discrete-time Markov chain, which we may call a *skeleton* of X .

First we address certain basics.

(3) Definition. The **transition probability** $p_{ij}(s, t)$ of the Markov chain X is defined to be

$$p_{ij}(s, t) = \mathbb{P}(X(t) = j \mid X(s) = i) \quad \text{for } s \leq t.$$

The chain is called **homogeneous** if $p_{ij}(s, t) = p_{ij}(0, t - s)$ for all i, j, s, t , and we write $p_{ij}(t - s)$ for $p_{ij}(s, t)$.

Henceforth we suppose that X is a homogeneous Markov chain, and we write $\mathbf{P}_t$ for the $|S| \times |S|$ matrix with entries $p_{ij}(t)$. The family $\{\mathbf{P}_t : t \geq 0\}$ is called the *transition semigroup* of the chain.

(4) Theorem. The family $\{\mathbf{P}_t : t \geq 0\}$ is a stochastic semigroup; that is, it satisfies the following:

- (a) $\mathbf{P}_0 = \mathbf{I}$, the identity matrix,
- (b) $\mathbf{P}_t$ is stochastic, that is $\mathbf{P}_t$ has non-negative entries and row sums 1,
- (c) the Chapman–Kolmogorov equations, $\mathbf{P}_{s+t} = \mathbf{P}_s \mathbf{P}_t$ if $s, t \geq 0$.

The Chapman–Kolmogorov equations of part (c) are also known as the *semigroup property*.

Proof. Part (a) is obvious.

(b) With $\mathbf{1}$ a row vector of ones, we have that

$$(\mathbf{P}_t \mathbf{1}')_i = \sum_j p_{ij}(t) = \mathbb{P}\left(\bigcup_j \{X(t) = j\} \mid X(0) = i\right) = 1.$$

(c) Using the Markov property,

$$\begin{aligned} p_{ij}(s+t) &= \mathbb{P}(X(s+t) = j \mid X(0) = i) \\ &= \sum_k \mathbb{P}(X(s+t) = j \mid X(s) = k, X(0) = i) \mathbb{P}(X(s) = k \mid X(0) = i) \\ &= \sum_k p_{ik}(s) p_{kj}(t) \quad \text{as for Theorem (6.1.7).} \quad \blacksquare \end{aligned}$$

As before, the evolution of $X(t)$ is specified by the stochastic semigroup $\{\mathbf{P}_t\}$ and the distribution of $X(0)$. Most questions about X can be rephrased in terms of these matrices and their properties.

Let X be a continuous-time Markov chain. We write $\mathbb{P}_i$ for the probability measure conditional on starting in the state $X(0) = i$, and $\mathbb{E}_i$ for the corresponding expectation. A certain amount of measure theory is needed in the rigorous study of continuous-time Markov chain. We shall avoid that here by presenting arguments that are partly intuitive. A greater degree of rigour may be found in Norris 1997.

There is a more general form of the Markov property than (1) above. For $t \geq 0$, we call an event A_t *t-historical* if it is given in terms of the past history $\{X(s) : s < t\}$, and *t-future* if it is given in terms of the future $\{X(s) : s > t\}$. We call the random time T a *stopping time* for the process X if, for all $t \geq 0$, the indicator function of the event $\{T \leq t\}$ is a function of the family $\{X(s) : s \leq t\}$.

(5) Theorem.

(a) **Extended Markov property.** Let $t > 0$, and let H_t be a *t-historical* event and F_t a *t-future* event. Then

$$\mathbb{P}_i(F_t \mid X(t) = j, H_t) = \mathbb{P}_j(F_t), \quad i, j \in S.$$

(b) **Strong Markov property.** Let T be a *stopping time* for X , and let H_T (respectively, F_T) be a *T-historical* (respectively, *T-future*) event. Then

$$\mathbb{P}_i(F_T \mid T < \infty, X(T) = j, H_T) = \mathbb{P}_j(F_T), \quad i, j \in S.$$

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The proof is omitted (see Norris 1997, p. 227), though diligent readers may adapt the ideas of the proof of Theorem (6.8.21).

We are ready to describe the architecture of the chain X . Let $X(0) = i$, and let $U_0 = \inf\{t : X(t) \neq i\}$ be the time of the first change of value. Since the chain is right-continuous, we have $U_0 > 0$. The chain evolves in time in the manner of the next two theorems.

(6) Theorem. The random variable U_0 has the exponential distribution with parameter denoted g_i and satisfying $g_i \in [0, \infty)$.

Proof. By the assumption of right-continuity, $\mathbb{P}_i(U_0 > 0) = 1$. By the extended Markov property at time s , and the homogeneity of the chain.

$$\mathbb{P}_i(U_0 > s + t \mid U_0 > s) = \mathbb{P}_i(U_0 > t),$$

so that $Q(t) := \mathbb{P}_i(U_0 > t)$ satisfies $Q(s + t) = Q(s)Q(t)$ for $s, t \geq 0$. By the lack-of-memory property (see Problem (4.14.15)), U_0 has the exponential distribution with some parameter g_i . The solution $\mathbb{P}_i(U_0 = \infty) = 1$ corresponds to the case $g_i = 0$. ■

Let the (Markov) transition matrix $\mathbf{Y} = (y_{ij} : i, j \in S)$ be given as follows. If $g_i = 0$, we set $y_{ij} = \delta_{ij}$. If $g_i > 0$, we set

$$y_{ij} = \mathbb{P}_i(X(U_0) = j), \quad j \in S.$$

Key to the analysis of Markov chains is the matrix $\mathbf{G} = (g_{ij} : i, j \in S)$, called the *generator* of the chain†, and defined by

$$(7) \quad g_{ij} = \begin{cases} g_i y_{i,j} & \text{if } j \neq i, \\ -g_i & \text{if } j = i. \end{cases}$$

Since $\mathbf{Y}$ has row sums 1, $\mathbf{G}$ has row sums 0, and so

$$(8) \quad \sum_{j: j \neq i} g_{ij} = g_i, \quad i \in S.$$

The element g_{ij} may be interpreted as the ‘intensity’ of jumps from i to j , that is,

$$(9) \quad \mathbb{P}(X(t + h) = j \mid X(t) = i) = g_{ij}h + o(h), \quad j \neq i.$$

This is easily proved when S is finite (*exercise*), but needs be treated with care when S is infinite. We note for later convenience that

$$(10) \quad g_{ij} = g_i(y_{ij} - \delta_{ij}), \quad i, j \in S.$$

(11) Theorem. Let $X(0) = i$ where $g_i > 0$.

- (a) The random variable $X(U_0)$ is independent of U_0 .
- (b) Write $X^*(s) = X(U_0 + s)$ for $s \geq 0$. Conditional on $X(U_0) = j$, the process X^* is a continuous-time Markov chain with the same transition probabilities as X , and with initial state $X^*(0) = j$. Furthermore, conditional on $X(U_0) = j$, X^* is independent of the family $\{X(u) : u \leq U_0\}$.

Proof. (a) By the extended Markov property at time u ,

$$\mathbb{P}_i(X(U_0) = j \mid U_0 > u) = \mathbb{P}_i(X(U_0) = j \mid U_0 > 0) = \mathbb{P}_i(X(U_0) = j),$$

so that $X(U_0)$ and U_0 are independent.

†Some writers use the historic notation q_{ij} in place of g_{ij} , and term the resulting matrix $\mathbf{Q}$ the ‘ Q -matrix’ of the process.

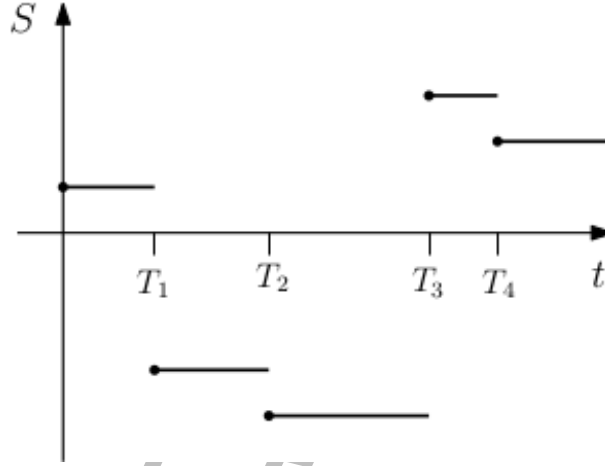


Figure 6.2. An illustration of the evolution of a Markov chain.

(b) This follows similarly from the strong Markov property at time T . ■

In summary, having started in state i , the chain X remains in i for an exponentially distributed time U_0 , with parameter g_i . It then jumps to a new state, j say, according to the transition matrix $\mathbf{Y}$. The chain then proceeds in a similar manner, starting in j , and otherwise independent of the history of X prior to time U_0 . Let T_m be the time of the m th change in value of X (with $T_0 = 0$), and let $U_m = T_{m+1} - T_m$. After the m th jump of X , the chain is constant for the exponentially-distributed time U_m called the m th ‘holding time’, before jumping to a new state according to $\mathbf{Y}$. A diagram of the evolution of X appears in Figure 6.2.

Here is another way of describing a continuous-time Markov chain. Let S be a countable state space, and let $\mathbf{Y} = (y_{ij} : i, j \in S)$ be the transition matrix of a discrete-time Markov chain Y taking values in S . We shall assume that $y_{ii} = 0$ for all $i \in S$; this is not an essential assumption, but recognizes the fact that jumps from any state i to itself will be invisible in continuous time. Let $g_i, i \in S$, be non-negative constants. We now construct a continuous-time chain X as follows. First, let $X(0) = Y_0$. After a holding time U_0 having the exponential distribution with parameter g_{Y_0} , the process jumps to the state Y_1 . After a further holding time U_1 having the exponential distribution with parameter g_{Y_1} , the chain jumps to Y_2 , and so on. The U_m are the *holding times* and the discrete-time chain Y is called the *jump chain*.

We argue more fully as follows. Conditional on the values (Y_n) of the chain Y , let $U_0, U_1, \dots$ be independent random variables having the respective exponential distributions with parameters $g_{Y_0}, g_{Y_1}, \dots$, and set $T_n = U_0 + U_1 + \dots + U_n$. We now define

$$(12) \quad X(t) = \begin{cases} Y_n & \text{if } T_n \leq t < T_{n+1} \text{ for some } n, \\ \infty & \text{otherwise.} \end{cases}$$

Equation (12) indicates a potential complication, namely the issue of ‘explosion’. Let $T_\infty = \lim_{n \rightarrow \infty} T_n$. It can be the case that T_∞ is finite with strictly positive probability, in which case $X(t)$ is not defined for $t \geq T_\infty$. If this holds, then X fails to satisfy the definition (2) of a Markov chain. This gap is filled by introducing a new state, which we choose to label ∞ , so that the new state space is $S' = S \cup \{\infty\}$. We define

$$(13) \quad X(t) = \infty \quad \text{for } t \geq T_\infty.$$

There are other ways of extending the time-domain of the process beyond $[0, T_\infty)$, and (4c) defines the so-called *minimal* process, since this process survives for the minimal time T_∞ . Recall the case of birth processes and Lemma (6.8.20). The process X is said to *explode* if $T_\infty < \infty$, and *explosion* is said to occur if, for some initial state, we have $\mathbb{P}(T_\infty < \infty) > 0$. Sufficient conditions for non-explosion are given in Theorem (17).

(14) Proposition. *Let X be a minimal process with transition probabilities $\{\mathbf{P}_t\}$. Then $\{\mathbf{P}_t\}$ satisfies the semigroup property $\mathbf{P}_{s+t} = \mathbf{P}_s \mathbf{P}_t$ for $s, t \geq 0$.*

Proof. The point is that an explosive process X may attain the additional state labelled ∞ , but it cannot return. Thus, for $i, j \in S$,

$$p_{ij}(s+t) = \sum_{k \in S} p_{ik}(s) p_{kj}(t), \quad i, j \in S,$$

as required. ■

Recall the generator $\mathbf{G} = (g_{ij} : i, j \in S)$ of the chain. This generator plays the role of the transition matrix $\mathbf{P}$ for discrete-time chains. We have that

$$(15) \quad \sum_j g_{ij} = 0 \quad \text{for all } i, \quad \text{or} \quad \mathbf{G}\mathbf{1}' = \mathbf{0}',$$

where $\mathbf{1}$ and $\mathbf{0}$ are row vectors of ones and zeros. A $S \times S$ matrix $\mathbf{K} = (k_{ij} : i, j \in S)$ is called a *generator* if $\mathbf{K}$ has non-negative off-diagonal elements and row sums 0.

(16) Example. Birth process (6.8.11). From the definition of this process, it is clear that

$$g_{ii} = -\lambda_i, \quad g_{i,i+1} = \lambda_i, \quad g_{ij} = 0 \quad \text{if } j < i \quad \text{or} \quad j > i+1.$$

Thus

$$\mathbf{G} = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & 0 & \dots \\ 0 & -\lambda_1 & \lambda_1 & 0 & 0 & \dots \\ 0 & 0 & -\lambda_2 & \lambda_2 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

It is useful to have verifiable conditions under which a chain does not explode.

(17) Theorem. *The chain X does not explode if any of the following three conditions holds:*

- (a) S is finite;
- (b) $\sup_i g_i < \infty$;
- (c) $X(0) = i$ where i is a persistent state for the jump chain Y .

Proof. First we prove that (b) suffices, noting in advance that (a) implies (b). Suppose that $g_i < \gamma < \infty$ for all i . The n th holding time U_n of the chain has the exponential distribution with parameter g_{Y_n} , where Y is the jump chain. If $g_{Y_n} > 0$, it is an easy exercise to show that $V_n = g_{Y_n} U_n$ has the exponential distribution with parameter 1. If $g_{Y_n} = 0$, then $U_n = \infty$ almost surely. Therefore,

$$\gamma T_\infty = \begin{cases} \infty & \text{if } g_{Y_n} = 0 \text{ for some } n, \\ \sum_{n=1}^{\infty} \gamma U_n \geq \sum_{n=1}^{\infty} V_n & \text{otherwise.} \end{cases}$$

It follows by Lemma (6.8.20) that the last sum is almost surely infinite; therefore, explosion does not occur.

Suppose now that (c) holds. If $g_i = 0$, then $X(t) = i$ for all t , and there is nothing to prove. Suppose that $g_i > 0$. Since $Y_0 = i$ and i is persistent for Y , there exists almost surely an infinity of times $N_0 < N_1 < \dots$ at which Y takes the value i . Now,

$$g_i T_\infty \geq \sum_{i=0}^{\infty} g_i U_{N_i},$$

and we may once again appeal to Lemma (6.8.20). ■

(18) Example. Let Y be a discrete-time chain with transition matrix $\mathbf{Y} = (y_{ij})$ satisfying $y_{ii} = 0$ for all $i \in S$, and let N be a Poisson process with intensity λ . We define X by $X(t) = Y_n$ if $T_n \leq t < T_{n+1}$ where T_n is the time of the n th arrival in the Poisson process (and $T_0 = 0$). The process X has transition semigroup $\mathbf{P}_t = (p_{ij}(t))$ given by

$$\begin{aligned} p_{ij}(t) &= \mathbb{P}_i(X(t) = j) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_i(X(t) = j, N(t) = n) \\ &= \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} e^{-\lambda t} \mathbb{P}_i(Y_n = j) = \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n (\mathbf{Y}^n)_{ij}}{n!}. \end{aligned}$$

We note that

$$e^{-\lambda t} \mathbf{I} = \sum_{n=0}^{\infty} \frac{(-\lambda t)^n}{n!} \mathbf{I}^n = \mathbf{I} e^{-\lambda t},$$

whence $\mathbf{P}_t = e^{\lambda t(\mathbf{Y} - \mathbf{I})}$. ●

(B) Irreducibility and persistence. We move on to the classification of states; this is slightly less of a chore than it was for discrete-time chains.

(19) Definition. The chain is called **irreducible** if, for any pair i, j of states, there exists $t > 0$ such that $p_{ij}(t) > 0$.

If the state space S is a singleton, it is trivial that the chain is irreducible. If $|S| \geq 2$, then irreducibility implies that $g_i > 0$ for $i \in S$.

(20) Theorem. If the chain X is irreducible, then $p_{i,j}(t) > 0$ for all $i, j \in S$ and $t > 0$.

Proof. Assume X is irreducible and let $i, j \in S$. Since $p_{i,j}(t) > 0$ for some $t > 0$, it is necessarily the case that $i \rightarrow j$ for the jump chain Y . On considering (7), we deduce that there exist distinct states $i_0, i_1, \dots, i_n$ with $i = i_0, j = i_n$ such that

$$g_{i_0, i_1} g_{i_1, i_2} \cdots g_{i_{n-1}, i_n} > 0.$$

If $g_{u,v} > 0$, then $g_u > 0$ so that

$$\begin{aligned} p_{u,v}(s) &\geq \mathbb{P}_u(U_0 \leq s, Y_1 = v, U_2 > s) \\ &= (1 - e^{-g_u s}) y_{u,v} e^{-g_v s} > 0, \quad s > 0. \end{aligned}$$

Therefore,

$$p_{i,j}(t) \geq p_{i_0,i_1}(t/n)p_{i_1,i_2}(t/n) \cdots p_{i_{n-1},i_n}(t/n) > 0,$$

as required. ■

The birth process is *not* irreducible, since it is non-decreasing. See Problem (6.15.16) for a condition for irreducibility in terms of the generator $\mathbf{G}$ of the chain.

The interplay between a continuous-time chain X and its jump chain Y provides a basic tool for the study of the former. We present just one example of this statement; others may be found in the exercises. We call⁴ the state i

- (21) *persistent* for X if $\mathbb{P}_i(\text{the set } \{t : X(t) = i\} \text{ is unbounded}) = 1$,
transient for X if $\mathbb{P}_i(\text{the set } \{t : X(t) = i\} \text{ is unbounded}) = 0$.

A persistent state i is called *positive* (or *non-null*⁵) if either $S = \{i\}$, or the earliest return time to i has finite mean, which is to say that the mean return time $m_i := \mathbb{E}_i R_i$ satisfies $m_i < \infty$, where $R_i := \inf\{t > U_0 : X(t) = i\}$. Positive persistence is important in the context of stationary distributions, to which we shall turn in the next section.

(22) Theorem. Consider the continuous-time chain X constructed above.

- (a) If $g_i = 0$, the state i is persistent for X .
- (b) Assume $g_i > 0$. State i is persistent for X if and only if it is persistent for the jump chain Y . Furthermore, i is persistent if the transition probabilities $p_{ii}(t) = \mathbb{P}_i(X(t) = i)$ satisfy $\int_0^\infty p_{ii}(t) dt = \infty$, and is transient otherwise.
- (c) Assume X is irreducible. Either every state is persistent or every state is transient.

Proof. It is trivial that i is persistent if $g_i = 0$, since then the chain X remains forever in the state i once it has first visited it.

Assume $g_i > 0$. If i is transient for the jump chain Y , there exists almost surely a last visit of y to i ; this implies the almost sure boundedness of the set $\{t : X(t) = i\}$, whence i is transient for X . Suppose i is persistent for Y , and $X(0) = i$. By Theorem (17c), the chain X does not explode. By the persistence of i , there exists almost surely an infinity of values n with $Y_n = i$. Since there is no explosion, the times T_n of these visits are unbounded, whence i is persistent for X .

Now, the integrand being positive, we may interchange limits to obtain

$$\begin{aligned} \int_0^\infty p_{ii}(t) dt &= \int_0^\infty \mathbb{E}_i(I_{\{X(t)=i\}}) dt \\ &= \mathbb{E}_i \left[\int_0^\infty I_{\{X(t)=i\}} dt \right] \\ &= \mathbb{E}_i \left[\sum_{n=0}^\infty U_n I_{\{Y_n=i\}} \right] \end{aligned}$$

where $\{U_n : n \geq 0\}$ are the holding times of X . The right side equals

$$\sum_{n=0}^\infty \mathbb{E}_i(U_0) y_{ii}(n) = \frac{1}{g_i} \sum_{n=0}^\infty y_{ii}(n)$$

⁴The word ‘recurrent’ is frequently used in place of persistent.

⁵A persistent state k is *null* if $m_k = \infty$.

where $y_{ii}(n)$ is the appropriate n -step transition probability of Y . By Corollary (6.2.4), the last sum diverges if and only if i is persistent for Y .

The final claim is trivial if $|S| = 1$, and we assume $|S| \geq 2$. Since X is assumed irreducible, we have that $g_i > 0$ for $i \in S$, and also the jump chain Y is irreducible. The claim now follows by the corresponding fact for irreducible discrete-time chains, namely Theorem (6.3.2). ■

Exercises for Section 6.9

1. Let $\lambda, \mu > 0$ and let X be a Markov chain on $\{1, 2\}$ with generator

$$\mathbf{G} = \begin{pmatrix} -\mu & \mu \\ \lambda & -\lambda \end{pmatrix}.$$

- (a) Write down the forward equations and solve them for the transition probabilities $p_{ij}(t)$, $i, j = 1, 2$.
 (b) Calculate $\mathbf{G}^n$ and hence find $\sum_{n=0}^{\infty} (t^n/n!) \mathbf{G}^n$. Compare your answer with that to part (a).
 (c) Solve the equation $\boldsymbol{\pi} \mathbf{G} = \mathbf{0}$ in order to find the stationary distribution. Verify that $p_{ij}(t) \rightarrow \pi_j$ as $t \rightarrow \infty$.
2. As a continuation of the previous exercise, find:
 (a) $\mathbb{P}(X(t) = 2 \mid X(0) = 1, X(3t) = 1)$,
 (b) $\mathbb{P}(X(t) = 2 \mid X(0) = 1, X(3t) = 1, X(4t) = 1)$.

3. Jobs arrive in a computer queue in the manner of a Poisson process with intensity λ . The central processor handles them one by one in the order of their arrival, and each has an exponentially distributed runtime with parameter μ , the runtimes of different jobs being independent of each other and of the arrival process. Let $X(t)$ be the number of jobs in the system (either running or waiting) at time t , where $X(0) = 0$. Explain why X is a Markov chain, and write down its generator. Show that a stationary distribution exists if and only if $\lambda < \mu$, and find it in this case.

4. **Pasta property.** Let $X = \{X(t) : t \geq 0\}$ be a Markov chain having stationary distribution $\boldsymbol{\pi}$. We may sample X at the times of a Poisson process: let N be a Poisson process with intensity λ , independent of X , and define $Y_n = X(T_n)$. Show that $Y = \{Y_n : n \geq 0\}$ is a discrete-time Markov chain with the same stationary distribution as X . (This exemplifies the ‘Pasta’ property: Poisson arrivals see time averages.)

[The full assumption of the independence of N and X is not necessary for the conclusion. It suffices that $\{N(s) : s \geq t\}$ be independent of $\{X(s) : s \leq t\}$, a property known as ‘lack of anticipation’. It is not even necessary that X be Markov; the Pasta property holds for many suitable ergodic processes.]

5. Let X be a continuous-time Markov chain with generator $\mathbf{G}$ satisfying $g_i = -g_{ii} > 0$ for all i . Let $H_A = \inf\{t \geq 0 : X(t) \in A\}$ be the hitting time of the set A of states, and let $\eta_j = \mathbb{P}_j(H_A < \infty)$ be the chance of ever reaching A from j . By using properties of the jump chain, which you may assume to be well behaved, show that $\sum_k g_{jk} \eta_k = 0$ for $j \notin A$.

6. In continuation of the preceding exercise, let $\mu_j = \mathbb{E}_j(H_A)$. Show that the vector $\boldsymbol{\mu}$ is the minimal non-negative solution of the equations

$$\mu_j = 0 \quad \text{if } j \in A, \quad 1 + \sum_{k \in S} g_{jk} \mu_k = 0 \quad \text{if } j \notin A.$$

7. Let X be a continuous-time Markov chain with transition probabilities $p_{ij}(t)$ and define $F_i = \inf\{t > T_1 : X(t) = i\}$ where T_1 is the time of the first jump of X . Show that, if $g_{ii} \neq 0$, then $\mathbb{P}_i(F_i < \infty) = 1$ if and only if i is persistent.

8. Let X be the simple symmetric random walk on the integers in continuous time, so that

$$p_{i,i+1}(h) = p_{i,i-1}(h) = \frac{1}{2}\lambda h + o(h).$$

Show that the walk is persistent. Let T be the time spent visiting m during an excursion from 0. Find the distribution of T .

9. Let i be a transient state of a continuous-time Markov chain X with $X(0) = i$. Show that the total time spent in state i has an exponential distribution.

10. Let X be an asymmetric simple random walk in continuous time on the non-negative integers with retention at 0, so that

$$p_{ij}(h) = \begin{cases} \lambda h + o(h) & \text{if } j = i + 1, i \geq 0, \\ \mu h + o(h) & \text{if } j = i - 1, i \geq 1. \end{cases}$$

Suppose that $X(0) = 0$ and $\lambda > \mu$. Show that the total time V_r spent in state r is exponentially distributed with parameter $\lambda - \mu$.

Assume now that $X(0)$ has some general distribution with probability generating function G . Find the expected amount of time spent at 0 in terms of G .

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13. (NEW) Let X be a Markov chain on the integers $\mathbb{Z}$ with generator satisfying $g_{i,i-1} = g_{i,i+1} = 2^i$ for $i \in \mathbb{Z}$, and $g_{i,j} = 0$ for other pairs (i, j) with $i \neq j$. Does X explode?

14. (NEW) **Growth with annihilation.** A population grows with the threat of annihilation. It is modelled as a Markov chain with generator $G = (g_{i,j})$ satisfying

$$\begin{aligned} g_{i,i+1} &= \frac{1}{i+2}, & i \geq 0, \\ g_{i,0} &= \frac{1}{(i+1)(i+2)}, & i \geq 0. \end{aligned}$$

the other off-diagonal elements of G being 0. Show that the chain is null persistent.

6.10 Kolmogorov equations and the limit theorem

As in the last section, X is a continuous-time Markov chain on the countable state space S , with transition semigroup $\{\mathbf{P}_t\}$ and generator $\mathbf{G}$. This section begins with an account of the relationship between $\{\mathbf{P}_t\}$ and $\mathbf{G}$.

(A) The forward and backward equations. The transition semigroup satisfies two differential equations, called the (Kolmogorov) forward and backward equations, which we introduce next.

Suppose for the moment that *the state space S is finite*, and let $X(0) = i$. By conditioning $X(t+h)$ on $X(t)$, we obtain

$$\begin{aligned} p_{ij}(t+h) &= \sum_k p_{ik}(t) p_{kj}(h) \\ &= p_{ij}(t)(1 + g_{jj}h) + \sum_{k: k \neq j} p_{ik}(t) g_{kj}h + o(h) \quad \text{by (6.9.9)} \\ &= p_{ij}(t) + h \sum_k p_{ik}(t) g_{kj} + o(h), \end{aligned}$$

giving that

$$(1) \quad \frac{1}{h} [p_{ij}(t+h) - p_{ij}(t)] \rightarrow \sum_k p_{ik}(t)g_{kj} = (\mathbf{P}_t \mathbf{G})_{ij} \quad \text{as } h \downarrow 0.$$

The above argument involves the passing of a limit through a summation. This is proper when S is finite, but requires justification when S is infinite.

With $\mathbf{P}'_t$ the matrix with entries $p'_{ij}(t)$, we write the *forward equations* (1) in the form

$$(2) \quad \mathbf{P}'_t = \mathbf{P}_t \mathbf{G}, \quad t \geq 0.$$

A similar argument, by conditioning $X(t+h)$ on $X(h)$, yields the *backward equation*

$$(3) \quad \mathbf{P}'_t = \mathbf{G} \mathbf{P}_t, \quad t \geq 0.$$

The common boundary condition for the forward and backward equations is $\mathbf{P}_0 = \mathbf{I}$.

Since S is finite, the forward and backward systems each forms a finite set of first-order linear differential equations which, subject to the boundary condition, has a unique solution. This solution may be expressed in the form

$$(4) \quad \mathbf{P}_t = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{G}^n,$$

the sum of powers of matrices (remember that $\mathbf{G}^0 = \mathbf{I}$). Equation (4) is deducible from (2) or (3) in very much the same way as we might show that the function of the single variable $p(t) = e^{gt}$ solves the differential equation $p'(t) = gp(t)$. The representation (4) for $\mathbf{P}_t$ is very useful and is usually written as

$$(5) \quad \mathbf{P}_t = e^{t\mathbf{G}} \quad \text{or} \quad \mathbf{P}_t = \exp(t\mathbf{G}),$$

where $e^{\mathbf{A}}$ is the natural abbreviation for $\sum_{n=0}^{\infty} (1/n!) \mathbf{A}^n$ whenever $\mathbf{A}$ is a square matrix. Some properties of matrix exponentials may be found in Exercise 1.

notation?

(6) Example. Consider a two-state chain X with $S = \{1, 2\}$; X jumps between 1 and 2 as time passes. There are two equivalent ways of describing the chain, depending on whether we specify $\mathbf{G}$ or we specify the holding times:

$$(a) \quad X \text{ has generator } \mathbf{G} = \begin{pmatrix} -\alpha & \alpha \\ \beta & -\beta \end{pmatrix},$$

(b) if the chain is in state 1 (respectively, 2), then it stays in this state for a length of time which is exponentially distributed with parameter α (respectively, β) before jumping to 2 (respectively, 1).

The equations of the forward system (2) take the form

$$p'_{11}(t) = -\alpha p_{11}(t) + \beta p_{12}(t),$$

and are easily solved to find the transition probabilities of the chain (*exercise*). ●

We now lift the assumption that S be finite. The above derivation requires an interchange of limits when S is infinite, and the story becomes more complicated. It turns out that the forward and backward equations may have more than one solution. Recall the term ‘minimal’ introduced in the discussion of explosion at (6.9.13).

(7) Theorem. *Let S be a countable set, and $\mathbf{G}$ a generator on S . Let X be a minimal Markov chain with generator $\mathbf{G}$.*

- (a) *The transition semigroup $\{\mathbf{P}_t\}$ of X is the minimal non-negative solution of the backward equation $\mathbf{P}'_t = \mathbf{G}\mathbf{P}_t$, subject to the boundary condition $\mathbf{P}_0 = \mathbf{I}$.*
- (b) *The semigroup $\{\mathbf{P}_t\}$ is also the minimal non-negative solution of the forward equation.*

In saying that $\{\mathbf{P}_t\}$ is the ‘minimal’ non-negative solution, we mean that $p_{ij}(t) \leq \pi_{ij}(t)$ for any other non-negative solution $\{\pi_t\}$, subject to the boundary condition. If there is no explosion, then $\sum_j p_{ij}(t) = 1$ for all $i \in S$ and $t > 0$, and (by Theorem (7)) its transition semigroup is the unique solution to both the forward and backward equations.

(8) Example. Birth process. Recall the birth process of Definition (6.8.11), and suppose that $\lambda_i > 0$ for all i . We saw in Section 6.8 that the condition $\sum_i \lambda_i^{-1} = \infty$ is necessary and sufficient for the forward and backward equations to have a unique solution. ●

Proof. We prove part (a), and refer the reader to Norris 1997, p. 100, for the proof of part (b). Write $p_{ij}(t)$ for the transition probabilities of the minimal chain X . By considering whether or not $J_0 > t$,

$$\begin{aligned} (9) \quad p_{ij}(t) &= \mathbb{P}_i(J_0 > t, X(t) = j) + \sum_{k: k \neq i} \mathbb{P}_i(J_0 \leq t, X(U_0) = k, X(t) = j) \\ &= \delta_{ij} e^{-g_i t} + \sum_{k: k \neq i} \int_0^t g_i e^{-g_i s} y_{ik} p_{kj}(t-s) ds, \end{aligned}$$

where δ_{ij} is the Kronecker delta. We change variables $u = t - s$ in the integral, and interchange the summation and integral (using a version of Fubini’s theorem), to obtain

$$e^{g_i t} p_{ij}(t) = \delta_{ij} + \int_0^t \sum_{k: k \neq i} g_i e^{g_i u} y_{ik} p_{kj}(u) du.$$

Now differentiate this equation (having checked the formalities), and substitute for the y_{ik} , to obtain the backward equation

$$(10) \quad p'_{ij}(t) = \sum_{k \in S} g_{ik} p_{kj}(t), \quad i \in S.$$

We show next that $\{\mathbf{P}_t\}$ is the minimal solution of the backward equation (10). Let $\{\widehat{\mathbf{P}}_t\}$ be a non-negative solution to (10) subject to the boundary condition. By reversing the steps from (9) to (10), we find that

$$(11) \quad \widehat{p}_{ij}(t) = \delta_{ij} e^{-g_i t} + \sum_{k: k \neq j} \int_0^t g_i e^{-g_i s} y_{ik} \widehat{p}_{kj}(t-s) ds.$$

We claim that

$$(12) \quad \hat{p}_{ij}(t) \geq \mathbb{P}_i(X(t) = j, J_n > t), \quad i, j \in S, t > 0, n \geq 0,$$

and we prove this by induction on n . Consider first the case $n = 0$. By (11),

$$\hat{p}_{ij}(t) \geq \delta_{ij}e^{-g_i t} = \mathbb{P}_i(X(t) = j, J_0 > t), \quad i, j \in S, t > 0.$$

Suppose (12) holds for $0 \leq n \leq N$ and consider the case $n = N + 1$. By the argument leading to (9),

$$\begin{aligned} \mathbb{P}_i(X(t) = j, J_{N+1} > t) \\ = \delta_{ij}e^{-g_i t} + \sum_{k: k \neq i} \int_0^t g_i e^{-g_i s} y_{ik} \mathbb{P}_k(X(t-s) = j, J_N > t-s) ds. \end{aligned}$$

On comparison with (11), we deduce (12) with $n = N + 1$. The induction step is complete.

The events on the right side of (12) are increasing in n , and furthermore $J_n \rightarrow T_\infty := \lim_{m \rightarrow \infty} T_m$, where T_m is the time of the m th jump of X . By the monotonicity of probability measures,

$$\begin{aligned} \hat{p}_{ij}(t) &\geq \lim_{n \rightarrow \infty} \mathbb{P}_i(X(t) = j, J_n > t) \\ &= \mathbb{P}_i(X(t) = j, T_\infty > t) = p_{ij}(t), \end{aligned}$$

as required. ■

(B) Stationary measures. Next we consider the limiting distribution of $X(t)$ as $t \rightarrow \infty$, in the non-explosive case. As in the discrete-time case, the large- t behaviour of $X(t)$ is closely bound up with the existence of stationary distributions and measures⁶. This definition may be compared with Definition (6.4.1). Recall that a *measure* $\mathbf{x}$ on S is a collection $\mathbf{x} = (x_i : i \in S)$ of non-negative reals satisfying $\mathbf{x} \neq \mathbf{0}$.

(13) Definition. Suppose X is irreducible and non-explosive with transition semigroup $\{\mathbf{P}_t\}$. The vector $\boldsymbol{\pi}$ is a **stationary measure** for X if $\boldsymbol{\pi}$ is a measure, and

$$\boldsymbol{\pi} = \boldsymbol{\pi} \mathbf{P}_t, \quad t \geq 0.$$

It is called a *stationary distribution* if, in addition, $\sum_j \pi_j = 1$.

A stationary measure $\boldsymbol{\pi}$ is said to be *unique* if, given any other stationary measure $\mathbf{v}$, there exists $\alpha > 0$ such that $\mathbf{v} = \alpha \boldsymbol{\pi}$. That is, uniqueness is subject to scalar multiplication.

If $X(0)$ has distribution $\boldsymbol{\mu}^{(0)}$ then the distribution $\boldsymbol{\mu}^{(t)}$ of $X(t)$ is given by

$$(14) \quad \boldsymbol{\mu}^{(t)} = \boldsymbol{\mu}^{(0)} \mathbf{P}_t.$$

If $\boldsymbol{\mu}^{(0)} = \boldsymbol{\mu}$, a stationary distribution, then $X(t)$ has distribution $\boldsymbol{\mu}$ for all t . For discrete-time chains we found stationary distributions by solving the equations $\boldsymbol{\pi} = \boldsymbol{\pi} \mathbf{P}$; the corresponding

⁶A stationary measure is also known as an *invariant measure*.

equations $\pi = \pi \mathbf{P}_t$ for continuous-time chains may seem complicated but they amount to a simple condition relating π and $\mathbf{G}$.

Here is the main theorem concerning the existence of stationary distributions. The reader is reminded of the first return time $R_i := \inf\{t > J_0 : X(t) = i\}$, with mean $m_i = \mathbb{E}_i R_i$.

(15) Theorem. *Let X be an irreducible Markov chain with state space S .*

- (a) *If some $k \in S$ is positive persistent, then there exists a unique stationary distribution π . Furthermore, π is the unique distribution that satisfies $\pi \mathbf{G} = \mathbf{0}$, and all states are positive persistent.*
- (b) *If X is non-explosive and there exists a distribution π satisfying $\pi \mathbf{G} = \mathbf{0}$, then: (i) all states are positive persistent, (ii) π is stationary, and (iii) $\pi_k = 1/(m_k g_k)$ for $k \in S$.*

The proof is preceded by a lemma.

(16) Lemma. *Let X be an irreducible chain, and suppose $|S| \geq 2$.*

- (a) *If $\mathbf{x}$ is a stationary measure, then $x_j > 0$ for all $j \in S$.*
- (b) *A measure $\mathbf{x}$ satisfies $\mathbf{x} \mathbf{G} = \mathbf{0}$ if and only if the measure $\mathbf{v} = (v_i : i \in S)$ given by $v_i = x_i g_i$ satisfies $\mathbf{v} \mathbf{Y} = \mathbf{v}$. If X is persistent, $\mathbf{x}$ is the unique measure (up to scalar multiplication) satisfying $\mathbf{x} \mathbf{G} = \mathbf{0}$.*
- (c) *Consider an irreducible persistent Markov chain X , and let $k \in S$. The measure $\mu(k) = (\mu_j(k) : j \in S)$ given by*

$$(17) \quad \mu_j(k) = \mathbb{E}_k \left(\int_0^{R_k} I_{\{X(s)=j\}} ds \right)$$

satisfies $\mu(k) \mathbf{G} = \mathbf{0}$ and is stationary for X .

Here as usual, I_A denotes the indicator function of the event A . The quantity $\mu_j(k)$ is to be viewed as the mean time spent in state j between two consecutive visits to k , and it plays the role of the vector $\rho(k)$ in the theory of discrete-time chains (see Lemma (6.4.5)).

Proof of Lemma (16). By irreducibility, $g_i > 0$ for $i \in S$.

(a) Since $\mathbf{x}$ is a measure, there exists $k \in S$ such that $x_k > 0$. Let $i \in S$. Since the chain is irreducible, by (6.9.20), $p_{ki}(1) > 0$, and, by stationarity, $x_i \geq x_k p_{ki}(1) > 0$.

(b) By (6.9.10),

$$\begin{aligned} (-\mathbf{v} + \mathbf{v} \mathbf{Y})_j &= -v_j + \sum_{i \in S} v_i y_{ij} \\ &= -x_j g_j + \sum_{i \in S} x_i (g_{ij} + g_i \delta_{ij}) = \mathbf{x} \mathbf{G}, \end{aligned}$$

as required. If X is persistent, by either Problem (6.15.7) or (6.15.8), $\mathbf{v}$ is the unique stationary measure of $\mathbf{Y}$ (up to scalar multiplication), so that $\mathbf{x}$ is the unique measure satisfying $\mathbf{x} \mathbf{G} = \mathbf{0}$.

(c) Since X is irreducible and persistent, by (6.9.17c) it is non-explosive. Let $\mu(k)$ be given as in (17). Then $\mu_j(k)$ is the mean time spent in state j before the first return time R_k to k . Such time is the sum of exponentially distributed random variables with parameter g_j , indexed by visits of the jump chain Y to j prior to R_k . Therefore,

$$(18) \quad \mu_j(k) = \frac{1}{g_j} \rho_j(k),$$

ADJUST space

where $\rho_j(k)$ is the mean number of visits by Y to j before its first return to its starting state k (recall (6.4.5)). By Lemma (6.4.7), $\rho(k) = (\rho_j(k) : j \in S)$ is a stationary measure for Y . By part (b) above, $\mu(k)$ satisfies $\mu(k)\mathbf{G} = \mathbf{0}$.

Next, we prove that $\mu(k)$ is a stationary measure in that $\mu(k)\mathbf{P}_t = \mu(k)$. By the strong Markov property (6.9.5b) at time R_k ,

$$\mathbb{E}_k \left(\int_0^t I_{\{X(s)=j\}} ds \right) = \mathbb{E}_k \left(\int_{R_k}^{R_k+t} I_{\{X(s)=j\}} ds \right), \quad t \geq 0.$$

Therefore, by displacing the integral over $[0, t]$,

$$\begin{aligned} \mu_j(k) &= \mathbb{E}_k \left(\int_t^{t+R_k} I_{\{X(s)=j\}} ds \right) = \mathbb{E}_k \left(\int_t^\infty I_{\{X(s)=j\}} I_{\{s-t < R_k\}} ds \right) \\ &= \int_0^\infty \mathbb{P}_k(X(u+t) = j, u < R_k) du \quad \text{where } u = s - t \\ &= \int_0^\infty \sum_{i \in S} \mathbb{P}_k(X(u) = i, u < R_k) p_{ij}(t) du \\ &= \sum_{i \in S} \mu_i(k) p_{ij}(t), \end{aligned}$$

as claimed. ■

Proof of Theorem (15). (a) Let k be a positive persistent state. By irreducibility and Theorem (6.9.22c), all states are persistent, and hence by Theorem (6.9.17c) X is non-explosive. Therefore, ‘no time is wasted’, which is to say that

$$(19) \quad R_k = \sum_j \int_0^{R_k} I_{\{X(s)=j\}} ds.$$

Let $\mu(k)$ be given as in (17). By (19),

$$\sum_j \mu_j(k) = \mathbb{E}_k R_k = m_k < \infty.$$

By Lemma (16c), $\pi := \mu(k)/m_k$ is a stationary distribution satisfying $\pi\mathbf{G} = \mathbf{0}$. Note, in particular, that $\pi_k = \mu_k(k)/m_k$. By (5c), $\mu_k(k) = 1/g_k$, so that

$$(20) \quad \pi_k = 1/(g_k m_k).$$

Let $i \in S$. We apply Lemma (16c) with k replaced by i to find that $\mu(i)$ satisfies $\mu(i)\mathbf{G} = \mathbf{0}$. By uniqueness, there exists $C_i \in (0, \infty)$ such that $\mu(i) = C_i \mu(k)$, whence

$$m_i = \sum_j \mu_j(i) = C_i \sum_j \mu_j(k) = C_i m_k.$$

Therefore, $m_i < \infty$, so that i is positive. Uniqueness follows from Lemma (16b).

(b) Suppose X is non-explosive and π is a distribution satisfying $\pi G = 0$. By Lemma (16b), $\nu = (\pi_i g_i : i \in S)$ satisfies $\nu Y = \nu$.

Let $k \in S$ and $\nu' = ((\pi_i g_i)/(\pi_k g_k) : i \in S)$, so that $\nu' Y = \nu'$ and $\nu'_k = 1$. By Problem (6.15.8), with $\rho(k)$ as in (18), we have $\rho(k) \leq \nu'$. Thus,

$$\rho_i(k) \leq \frac{\pi_i g_i}{\pi_k g_k}, \quad i \in S.$$

This we sum over i to obtain

$$m_k = \sum_i \mu_i(k) = \sum_i \frac{\rho_i(k)}{g_i} \leq \frac{1}{\pi_k g_k} < \infty,$$

proving that every k is positive persistent. It is here that we have used non-explosivity, and specifically equation (19).

Equation (20) was proved under the assumption that k is persistent. It holds, therefore, for every $k \in S$. ■

(21) Example. Unlike the situation in discrete time, there may exist a distribution π satisfying $\pi G = 0$, and yet the chain is transient. The reason is essentially that a stationary distribution is influenced by the transition matrix of the jump chain, whereas explosiveness is influenced by the holding times.

Let X be the chain with state space $S = \{0, 1, 2, \dots\}$ and generator

$$g_{ij} = \begin{cases} \lambda g_i & \text{if } j = i + 1, \\ \mu g_i & \text{if } j = i - 1 \geq 0, \\ 0 & \text{for other pairs } i \neq j, \end{cases}$$

where $\lambda, \mu > 0$, and $g_i > 0$ for $i \in S$. It is easily seen that the measure

$$\nu_i = \frac{1}{g_i} \left(\frac{\lambda}{\mu} \right)^i, \quad i \in S,$$

satisfies $\nu G = 0$. If $g_i \equiv 1$ and $\lambda/\mu < 1$, ν may be normalized to obtain a distribution π . This distribution is stationary for the chain. Suppose, in contrast, that $g_i = 2^i$ and $\lambda/\mu \in (1, 2)$. Then ν is summable, so it may be normalized to a distribution π . However, the jump chain is transient, and therefore X is transient also. By Theorem (15c), X is explosive. ●

(C) The limit theorem. We turn to the limit theorem for Markov chains, sometimes known as the ergodic theorem. What can be said about the limiting distribution of $X(t)$ as $t \rightarrow \infty$?

(22) Theorem. Let X be irreducible and non-explosive.

(a) If there exists a stationary distribution π then it is unique and

$$p_{ij}(t) \rightarrow \pi_j \quad \text{as } t \rightarrow \infty, \quad \text{for all } i \text{ and } j.$$

(b) If there is no stationary distribution then $p_{ij}(t) \rightarrow 0$ as $t \rightarrow \infty$, for all i and j .

The proof is preceded by a lemma concerning the continuity of transition probabilities.

(23) Lemma. Let X be a minimal Markov chain with transition semigroup $\{\mathbf{P}_t\}$ and generator $\mathbf{G}$. Then

$$|p_{ij}(t+u) - p_{ij}(t)| \leq 1 - e^{-g_i u}, \quad i, j \in S, \quad t, u \geq 0.$$

Proof. Since $\{\mathbf{P}_t\}$ has the semigroup property (6.9.14),

$$\begin{aligned} |p_{ij}(t+u) - p_{ij}(t)| &\leq \left| \sum_k p_{ik}(u) p_{kj}(t) - p_{ij}(t) \right| \\ &= \left| \left(\sum_{k: k \neq i} p_{ik}(u) p_{kj}(t) \right) - p_{ij}(t)[1 - p_{ii}(u)] \right| \\ &\leq 1 - p_{ii}(u) \\ &\leq \mathbb{P}_i(J_0 \leq u) = 1 - e^{-g_i u}, \end{aligned}$$

as required. ■

Proof of Theorem (22). We may assume that $|S| \geq 2$, whence $g_i > 0$ for $i \in S$. Fix $h > 0$ and let $Z_n = X(nh)$. Then $Z = \{Z_n\}$ is an irreducible, aperiodic, discrete-time Markov chain, called a *skeleton* of X . By the comments following Theorem (6.4.20),

$$(24) \quad p_{ij}(nh) \rightarrow l_{ij}^h \quad \text{as } n \rightarrow \infty,$$

where

$$(25) \quad l_{ij}^h = \begin{cases} \pi_j^h & \text{if } Z \text{ is non-null persistent,} \\ 0 & \text{otherwise,} \end{cases}$$

where, in the non-null persistent case, $\pi^h = (\pi_j^h : j \in S)$ is the unique stationary distribution of Z . We now fill in the gaps via the continuity of Lemma (23).

Let $i, j \in S$, $\epsilon > 0$, and pick $h > 0$ such that

$$1 - e^{-g_i h} \leq \frac{1}{2}\epsilon.$$

By (24), we may find N such that

$$|p_{ij}(nh) - l_{ij}^h| \leq \frac{1}{2}\epsilon, \quad n \geq N.$$

For $t > Nh$, let n be such that $nh \leq t < (n+1)h$. By Lemma (23),

$$|p_{ij}(t) - l_{ij}^h| \leq |p_{ij}(t) - p_{ij}(nh)| + |p_{ij}(nh) - l_{ij}^h| \leq \epsilon.$$

The limits follow.

In the positive persistent case, the uniqueness of π holds since any stationary distribution for X is also stationary for Z . ■

(26) Example (6) revisited. If $\alpha, \beta > 0$ and $S = \{1, 2\}$, then

$$\mathbf{G} = \begin{pmatrix} -\alpha & \alpha \\ \beta & -\beta \end{pmatrix}$$

is the generator of a stochastic semigroup $\{\mathbf{P}_t\}$ given by the following calculation. Diagonalize $\mathbf{G}$ to obtain $\mathbf{G} = \mathbf{B}\mathbf{A}\mathbf{B}^{-1}$ where

$$\mathbf{B} = \begin{pmatrix} \alpha & 1 \\ -\beta & 1 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} -(\alpha + \beta) & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore

$$\begin{aligned} \mathbf{P}_t &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{G}^n = \mathbf{B} \left(\sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{A}^n \right) \mathbf{B}^{-1} \\ &= \mathbf{B} \begin{pmatrix} h(t) & 0 \\ 0 & 1 \end{pmatrix} \mathbf{B}^{-1} \quad \text{since } \mathbf{A}^0 = \mathbf{I} \\ &= \frac{1}{\alpha + \beta} \begin{pmatrix} \alpha h(t) + \beta & \alpha[1 - h(t)] \\ \beta[1 - h(t)] & \alpha + \beta h(t) \end{pmatrix} \end{aligned}$$

where $h(t) = e^{-t(\alpha+\beta)}$. Let $t \rightarrow \infty$ to obtain

$$\mathbf{P}_t \rightarrow \begin{pmatrix} 1 - \rho & \rho \\ 1 - \rho & \rho \end{pmatrix} \quad \text{where } \rho = \frac{\alpha}{\alpha + \beta}$$

and so

$$\mathbb{P}(X(t) = i) \rightarrow \begin{cases} 1 - \rho & \text{if } i = 1, \\ \rho & \text{if } i = 2, \end{cases}$$

irrespective of the initial distribution of $X(0)$. This shows that $\pi = (1 - \rho, \rho)$ is the limiting distribution. Check that $\pi \mathbf{G} = \mathbf{0}$. The method of Example (6) provides an alternative and easier route to these results. ●

Exercises for Section 6.10

1.

new

Let $N < \infty$, and let $\mathcal{Q}$ be the space of $N \times N$ matrices with real entries, with norm

$$\|Q\| = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{|Q\mathbf{x}|}{|\mathbf{x}|}, \quad Q \in \mathcal{Q}.$$

(a) Show that

$$\|Q_1 + Q_2\| \leq \|Q_1\| + \|Q_2\|, \quad \|Q_1 Q_2\| \leq \|Q_1\| \cdot \|Q_2\|.$$

(b) Show that $E_n := \sum_{m=0}^n Q^m / m!$ converges with respect to $\|\cdot\|$ to a limit which we denote $E = E(Q)$.

(c) Show that $E(Q_1 + Q_2) = E(Q_1)E(Q_2)$ if Q_1 and Q_2 commute.

2. Let Y be an irreducible discrete-time Markov chain on a countably infinite state space S , having transition matrix $\mathbf{Y} = (y_{ij})$ satisfying $y_{ii} = 0$ for all states i , and with stationary distribution $\mathbf{v}$. Construct a continuous-time process X on S for which Y is the jump chain, such that X has no stationary distribution.

Old one from previous section. Notation changed

3.

more...